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Deep Learning in Sinhala Sign Language Recognition: A Systematic Decade Review

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***Abstract*—The paper reviews the progress in Sinhala Sign Language recognition (SSL) over the last decade, tracing the techniques from early image processing to advanced deep learning models, indicating methodologies involved, datasets used, and challenges. The paper shows that the evolution has progressed from basic classifiers to sophisticated machine learning techniques that integrate CNNs, RNNs, and transformer architectures. It also presents the implementation of multimodal systems using hand gestures, facial expressions, and body postures. The review identifies the limitations of current recognition in SSL, such as scarce complete datasets, technical difficulties, and a lack of robust continuous recognition. Finally, it points out future research directions, stressing that further work must focus on growing datasets, making lightweight models for edge devices, and collaboration for further growth in SSL recognition technology.**

Keywords*—Sri Lankan Sign Language Recognition, CNN, LSTM, RNN, Gesture Recognition

I. Introduction

According to a World Health Organization report [50], over 5% of the global population, including 34 million children, requires rehabilitation for hearing loss, with projections indicating that by 2050, 2.5 billion people will experience some degree of hearing loss, and 700 million will rely on hearing aids. The issue is particularly pressing in Sri Lanka, where a study in 2023 [22] highlighted the marginalization and under-support of deaf and hard-of-hearing individuals, emphasizing the urgent need for reform. Sign languages, such as Sinhala Sign Language (SSL), play a crucial role in communication for these communities.

According to [26], SSL consists of unique gestures, hand shapes, orientations, movements, and facial expressions representing words and phrases. Despite significant advancements in automated sign language recognition through sensors, computer vision, machine learning, and natural language processing, SSL recognition remains challenging due to its specific gestures and the lack of extensive annotated datasets compared to more widely studied sign languages.

SSL recognition research is relatively recent, beginning in the early 1990s compared to other languages like British Sign Language [1] and American Sign Language [30] which root back to the 1970s. Early efforts focused on recognizing static signs using basic image processing techniques, with a study in 1993 [49] documenting the use of Neural Networks for ASL translation. Initial studies, done by [42], analyzed hand shapes from still images. Meanwhile, early Japanese sign language research used sensor-based methods, including the wearable "DataGlove" to capture hand shape and position data [27].

The late 1990s and early 2000s brought significant advancements in computer vision and machine learning, enhancing sign language recognition systems. A 1997 study [9] introduced a video-based system for recognizing 262 signs from the Sign Language of the Netherlands, achieving up to 94% accuracy for isolated signs using Hidden Markov Models (HMMs). HMMs enabled effective modeling of dynamic gestures. In 2012 [21], a real-time dynamic hand gesture system using a depth camera identified 12 ASL gestures. A 2015 study on Indian Sign Language (ISL) recognition using a Leap Motion sensor achieved 88.39% and 90.32% accuracies with Euclidean Distance and Cosine similarity methods, though it struggled with non-manual signs [25].

SSL recognition research gained momentum in the late 2000s, facing challenges such as the lack of annotated datasets. The 2010s brought a revolution with deep learning techniques, including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), improving feature extraction and temporal modeling. Recent years have seen increased interest in applying advanced models like Long Short-Term Memory (LSTM) networks and Transformer architectures to SSL recognition, showing promising results in accuracy and generalization across different signers and environments. The research trend in SSL can be seen in Figure 1. (rough estimate based on publicly available data).

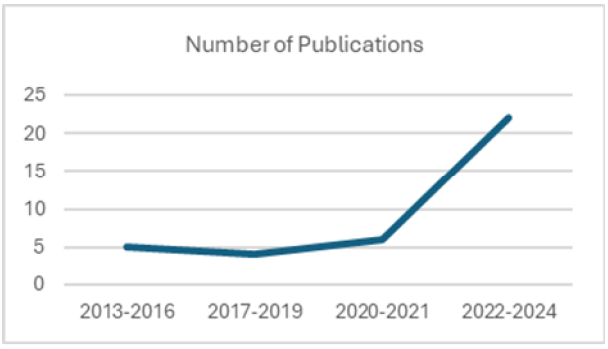


Fig. 1. Sinhala Sign Language Research Trend

This paper reviews Sinhala Sign Language recognition, aiming to fill the literature gap by introducing innovative methodologies and evaluation strategies. It provides a comprehensive review of related works, discusses dataset

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development and preprocessing techniques, presents core contributions of proposed models and algorithms, details about implementation and performance evaluations, and discusses the implications of the findings on potential applications and future research directions in this domain.

II. Methodology

To guarantee thorough coverage and reduce bias, this systematic review employed an organized methodology to examine Sri Lankan Sign Language (SSL) recognition research conducted during the previous ten years. Additionally, the history of sign language from 1970 to 2024 was covered to understand the origins and learn from both successful and unsuccessful attempts.

A. Search Pipeline

The literature search was conducted using the following strategy.

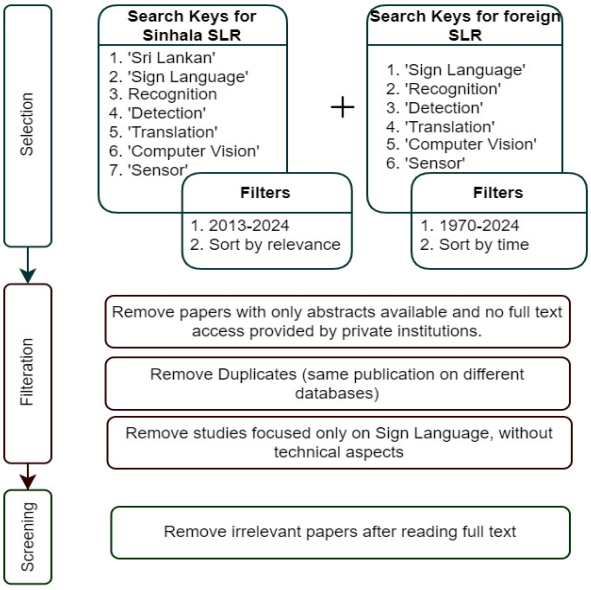


Fig. 2. Literature Search Methodology

The above diagram (Figure 2) outlines a systematic methodology for conducting this literature review on sign language recognition, focusing on both Sinhala Sign Language (SLR) and foreign sign languages. The process involves selection, filtration, and screening stages. The selection phase uses search keys and date filters, while the filtration stage removes limited access, duplicates, and technical studies. The screening phase removes irrelevant papers after a full-text review, ensuring a comprehensive and relevant literature collection.

B. Data Extraction and Analysis

The literature review used a holistic approach to extract and analyze data on recognition, including author details, year of publication, research objectives, methodologies, dataset descriptions, principal findings, performance metrics, and limitations. The data was tabulated in a spreadsheet for comparison and detailed analysis. The studies were organized by time, methodological approach, and recognition type. Topic-based analysis helped identify recurrent themes, emerging trends, and gaps in existing literature. This systematic analysis formed the basis for the synthesis presented in sections four and five, providing a structured overview of the development and current state of the field.

C. Quality Assessment

The quality assessment of included studies was conducted using the Critical Appraisal Skills Program checklist [2]. This assessment focused on key criteria including the clarity of research objectives, appropriateness of methodology, rigor of data analysis, validity of conclusions, and relevance to SSL recognition. This methodical approach to quality evaluation ensured a systematic and comprehensive review of the SSL recognition literature, thereby establishing a robust foundation for the analysis and conclusions presented in the subsequent sections of the review. The search results of foreign sign language will be discussed in depth in the next section.

III. Overview of Foreign Sign Language

Sign Language Recognition (SLR) is a vastly intense research area of both computer vision and natural language processing with a standing history, as highlighted in the previous section. Sign languages are recognized as languages by linguists like any other language, written and spoken around the world so far. It deals with translating sign language into text or spoken language, the recognition of communication aid to use by the deaf and hard-of-hearing. The aim here is to create systems that can efficiently recognize and interpret the complicated gestures that are part of sign language.

A. Methodology

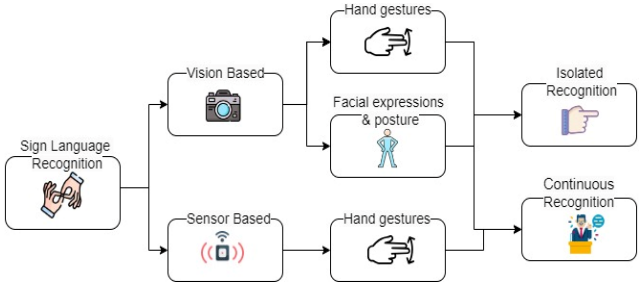


Fig. 3. Sign Language Recognition Category Map

As depicted in Figure 3, SLR systems are usually categorized into two main categories: vision-based and sensor-based, as illustrated in the above diagram and according to [23].

1) Vision-based: It utilizes cameras for tracking hand and body gestures. Being non-intrusive, this approach is the most used because of its feasibility of real-time applications. Variation in lighting conditions, background noise, and occlusions are some of the challenges to be addressed.

2) Sensor-based: It consists of gloves fitted with sensors to track the movements of hands. These systems are highly accurate but intrusive and comparatively expensive, hence less pragmatic for use in everyday life.

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Furthermore, Sign language can be categorized into isolated and continuous forms, with manual components like hand gestures representing isolated signs and non-manual components like facial expressions and body posture representing continuous signs. Isolated sign language recognition (SLR) focuses on identifying individual signs.

B. Recent Advances

Most of the recent developments in SLR have come through the application of various deep learning techniques: CNN, RNNs, and Transformers, which have greatly improved recognition in sign language [45]. Notable amongst these are word-level SLR, which involves isolating recognition at single words in sign language, using huge datasets, and contextual attention mechanisms that further increase the accuracy of recognition by considering the context in which signs occur. According to [31], Transfer learning has also emerged as another effective technique where pre-trained models are applied to new data sets of the sign language. This step mostly helped to increase the accuracy even with small datasets, and recent papers showed this approach. These developments represent a new leap forward in the field of SLR, paving the path toward more accurate and efficient systems for sign language interpretation.

C. Datasets and Benchmarks

According to [31], several datasets have been developed to facilitate SLR research, including:

TABLE I. BENCHMARKS FOR SIGN LANGUAGE RECOGNITION

Dataset	***Benchmark Description***
WLASL	(Word-Level American Sign Language) A large-scale video dataset for recognizing 2000 individual words [47]. The NLA-SLR model has achieved the highest accuracy of 61.26 [51]
AUTSL	(Ankara University Turkish Sign Language Dataset) Provides a comprehensive set of sign language videos for training and evaluation. It contains 226 signs that are performed by 43 different signers and 38,336 video samples in total [40]. The Spatio-Temporal Features (STF) +LSTM approach has achieved the highest recognition rate of 98.56% [39].
MS-ASL	(Microsoft American Sign Language) A large-scale sign language data set of over 25,000 annotated videos is proposed, covering 1000 signs in real-life recording conditions. The data set also proposes I3D architecture as a powerful and suitable architecture for sign language recognition, outperforming current methods [18]. The SignBert+ framework has achieved the highest accuracy of 73.71% [14].

Table 1 above highlights a few key datasets that are essential for comparing various SLR models and assessing how well they function in practical settings. The provided resource contains many more recent datasets [31].

IV. SINHALA SIGN LANGUAGE

As mentioned in the introduction, SSL is the primary mode of communication for Sri Lanka's deaf and hard-of-hearing communities. Unlike widely researched sign languages such as American Sign Language (ASL) and British Sign Language (BSL), SSL is relatively unknown, posing unique challenges and opportunities for researchers. Over the last decade, research on SSL recognition has advanced significantly, using a variety of technologies and methodologies to improve communication for Sri Lanka's deaf and hard of hearing communities. The evolution state can be seen in the diagram below, where the relevant literature and methodologies will be discussed.

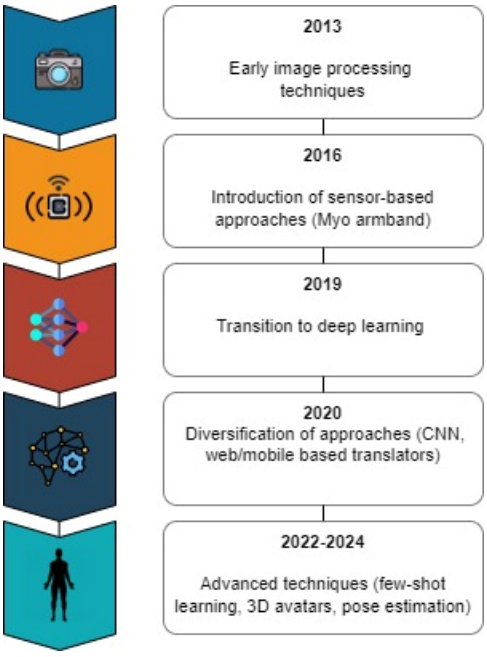


Fig. 4. Evolution of Sinhala Sign Language

A. Early Approaches (2013-2016)

As shown in Figure 4, initial studies focused on image processing techniques and basic classifiers, rather than Machine Learning techniques. The research [12] proposed an image-based system using skin color filtering and centroid finding methods, achieving 93.3% accuracy for 15 static gestures. [48] developed a computer vision approach utilizing morphological operations and contour matching. This was achieved through a Kinect camera [36] and OpenCv. Similarly, [5] introduced a system using a Kinect sensor and Dynamic Time Warping for classifying both static and dynamic gestures. Advancements in technology led to the exploration of sensor-based approaches. [24] proposed a wearable solution using the Myo armband to capture hand and arm movements for sign recognition. [43] developed a portable tool using flex sensors and a 3D accelerometer for gesture recognition.

B. Transition to Machine Learning (2017-2019)

The field saw a shift towards more sophisticated machine learning techniques. [32] employed an Inception model to identify 27 static SSL gestures, creating a custom dataset of 27,000 images. [8] was a part of the first efforts to utilize a Myo armband and machine learning models for sentence-level continuous signing recognition.

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C. Expansion and Diversification (2020-2021)

This period saw a lot of approaches and applications in SSL research. [4] trained a Convolutional Neural Network (CNN) to learn 12 SSL gesture types, employing a one-time segmentation process for different skin colors through a Gaussian Mixture Model. This approach achieved a higher accuracy rate for recognition and added both Tamil and Sinhala language translation capabilities, along with the ability to train on user-provided data for the recognition process. [19] presented EasyTalk, a web-based sign language translator capable of translating SSL into text and audio formats and verbal language into SSL. The model demonstrated robustness in detecting signs in complex backgrounds, bright and dark environments, and flipping hands.

[15] developed a vision-based SSL interpreter optimized for real-time use on portable devices. This interpreter facilitated conversations between hearing-impaired and non-signing individuals using both static and dynamic signs. Key techniques included skin segmentation, contour extraction, hand detection, and the mean shift algorithm to isolate the signing hand. A CNN model was used to extract and classify static and dynamic signs, achieving average accuracies of 99.5% and 81.2%, respectively. [37] proposed a mobile solution employing a Region-Based Convolutional Neural Network for object detection, aiming to reduce the gap in real-time sign identification and interpretation using Natural Language Processing (NLP). [33] utilized Artificial Neural Networks (ANN) and Support Vector Machines (SVM) for translating 24 static SSL alphabet words and numbers. Although their specific link is not always accessible, their contribution remains noteworthy in the context of SSL research expansion during this period.

D. Recent Advancements (2022-2024)

In recent years, research in Sinhala Sign Language (SSL) recognition and translation has seen significant progress, focusing on improving accuracy, expanding datasets, and exploring novel approaches. This summary highlights key developments from 2022 to 2024. Several studies have leveraged advanced machine learning techniques to enhance SSL recognition. For instance, [46] developed an ensemble classifier using Logistic Regression, K-Nearest Neighbors, and Random Forest for static sign recognition, achieving competitive accuracy with a small dataset of 122 images. [13] compared various algorithms using Google MediaPipe Hand Pose landmarks, reporting that Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), and CNN-LSTM models achieved over 95% accuracy. [34] also used a LSTM model enhancing location-specific dynamic sign recognition, with an accuracy of 98.89%. Additionally, [11] proposed a multi-modal Deep Learning approach for sentence-level sign gesture recognition, incorporating hand and lip movements. Their model achieved a best Word Error Rate (WER) of 12.70, utilizing ResNet, transformer encoders, and CTC alignment.

Addressing the challenges of low-resource languages, [17] introduced ProtoSign, a new architecture for few-shot learning in SSL. This approach, along with the SSL-50 dataset, demonstrated effectiveness in low-resource settings across multiple datasets. Researchers have also explored various input modalities and sensor technologies. [7] investigated the use of SSL in sign language translation, incorporating emotion recognition and gesture recognition using deep learning and 3D modeling techniques. [28] developed a wearable, sensor-based gesture recognition glove coupled with an LSTM model, achieving 83% categorical accuracy with a 320ms inference time. Furthermore, studies utilized Leap Motion technology and Natural Language Processing for both static and dynamic SSL signs, reporting over 99% [38] accuracy in user-independent testing, and another similar study focusing on a comparative study [20] achieved 89.2% accuracy.

Several projects aimed to create complete communication systems. [10] focused on developing a system to enhance interactions for individuals with disabilities, including deaf-blind individuals, providing textual, speech, and braille output. [16] proposed a system for children incorporating a learning platform, dynamic sign detection, audio/video to sign conversion, and vocal training, where similar work [35] had been done specifically for children with cochlear implants. Moreover, [6] combined static sign recognition using MediaPipe and SVM algorithms with natural language processing to produce meaningful sentences, achieving 99.67% accuracy. [29] proposed a sign language detector using various machine learning algorithms with human pose estimation, focusing on upper-body stance landmarks and achieving an 86.75% success rate.

V. Literature Discussion

In reviewing recent advancements in Sinhala Sign Language (SSL) recognition and translation from 2022 to 2024, notable progress includes work by [13] over 95% accuracy using LSTM, CNN, and CNN-LSTM models. [11] achieved a 12.70 Word Error Rate in sentence-level recognition with ResNet and transformer encoders. ProtoSign by [17], effectively addressed low-resource settings. [28] sensor-based glove achieved 83% accuracy with a 320ms inference time, and [38] reported over 99% accuracy using Leap Motion technology and NLP.

Sign Language Systems (SSL) have made significant progress in Sri Lanka's deaf and hard of hearing community, but there are still limitations and future directions. These include limited datasets, technical constraints, and difficulty with continuous recognition and context-dependent signs. Most studies lack proper validation or benchmarking, making accuracy claims questionable. The lack of public datasets for comparison further complicates the situation. Research has not advanced beyond initial proof-of-concept stages, indicating a need for more ambitious efforts. Additionally, there is a lack of focus on lightweight models suitable for edge devices, which limits their applicability in real-world scenarios where processing occurs on mobile or low-power devices. Overall, there is a need for more ambitious and comprehensive efforts in sign language recognition systems. To address these limitations, several directions for future research are proposed in table 2.

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TABLE II. FUTURE DIRECTIONS

Direction	*Details*
Data Expansion	Future work should focus on expanding data collection to increase the volume and diversity of sign language datasets. This includes covering a broader range of signs, contexts, and signing styles. Establishing standards for data collection and annotation is crucial to ensure consistency and quality across studies.
Collaborate	Encouraging collaboration between researchers, institutions, and the deaf community is essential for creating comprehensive and representative datasets. This collaborative approach can lead to more robust and culturally sensitive recognition systems
Recognition Techniques	Researchers should work on incorporating facial expression recognition and body posture analysis, developing methods for continuous sign language recognition, and improving context-aware interpretation of signs.
Lightweight Models	Future systems should improve real-time processing capabilities and optimize models for low-capacity devices, ensuring wider accessibility and usability. Lightweight models designed for edge computing will enable deployment on mobile devices and other resource-constrained platforms, making sign language recognition more practical.
Multi-modality	Integrating multiple sensor types and developing adaptive learning mechanisms can lead to more robust and personalized recognition systems.
Cross-lingual Recognition	Expanding research to cover multiple sign languages and exploring transfer learning techniques between different sign languages is an important future direction.
Ethical Concerns	Addressing privacy concerns related to data collection and ensuring inclusivity in system design should be prioritized in future research, especially in computer vision approaches where the user needs to always keep the camera on.
Validation Protocols	Developing standardized benchmarking datasets and validation protocols is crucial for fair comparison of different approaches and for establishing the real-world effectiveness.

By prioritizing these future directions, with a particular emphasis on publishing the created custom datasets, benchmarking/validation, and the development of lightweight models for edge devices, the field of sign language recognition can advance significantly. Proper benchmarking and validation ensure that research results are trustworthy, while lightweight models enhance accessibility and usability. This combined focus will help create more accurate, robust, and practical systems that genuinely meet the needs of the deaf community in diverse real-world settings.

VI. DATASETS FOR SSL

One of the basic requirements for both training and evaluation of the sign language recognition system is datasets. Sinhala Sign Language belongs to the category of low-resource languages; thus, it has been rather difficult to construct full-fledged datasets. A notable trend in literature reveals that while many researchers have created their own datasets, few have taken steps to make these publicly available. However, recent efforts have been made to create and improve datasets specifically tailored for SSL recognition.

Currently, there are only three publicly available datasets for SSL:

1) This dataset by [3], available on Kaggle, includes various sign gestures used in SSL. It aims to provide a foundational resource for researchers and developers working on SSL recognition projects. The dataset contains images and video sequences of sign language gestures, which are essential for training machine learning models.

2) The dataset by [44] focuses on SSL alphabets but is limited in scope. It contains only 14 alphabet letters, with 50 data points for each letter. While it provides a starting point for alphabet recognition, its limited coverage may restrict its usefulness in comprehensive SSL recognition systems

3) The dataset by [41], named "SSL-50" dataset offers a broader range of SSL signs, covering 50 different classes. However, it is also limited in its depth, providing only 20 videos per class. This data set can be useful for initial model training and testing but may not be sufficient for developing robust, large-scale SSL recognition systems.

While these datasets represent important steps towards facilitating SSL recognition research, they highlight the ongoing need for more comprehensive and diverse SSL datasets. The limited scope and depth of existing resources underscore the challenges faced by researchers in this field and emphasize the importance of continued efforts to develop and share more extensive SSL datasets.

VII. CONCLUSION

In conclusion, this study makes a significant contribution to the field of sign language recognition by addressing a critical gap in the research on Sinhala Sign Language, a relatively unexplored area in comparison to other sign languages. While existing literature has primarily focused on widely used sign language, this research introduces methodologies and provides comprehensive evaluations that are specifically designed for the Sinhala context. Fulfilling the identified research gaps along with the future directions offers a robust foundation for further research and development in this underrepresented domain. The findings highlight the unique challenges and opportunities associated with Sinhala Sign Language recognition, paving the way for future advancements in creating more inclusive and accessible communication technologies.

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